Stochastic Simulation - Assignment B
Ethereum Gas Market Simulation

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1 Introduction

In this report, we simulate the Ethereum mempool under the EIP-1559 mechanism with a dynamic base fee. A mempool is a queue of unconfirmed transactions. The validator selects eligible transactions in descending order of tip. A transaction is included if its gas demand fits within the remaining block gas budget. Otherwise, it is skipped, and the algorithm continues with the next eligible transaction. The base fee is adjusted every block. At time $t = 0$, the mempool is empty, and the base fee $b_0 = 10$ Gwei. If a transaction stays in the mempool for more than 300 seconds, then it expires and is thus removed from the mempool.

2 Event types

The simulation is implemented as a discrete-event simulation with three main event types: transaction arrival, block production, and transaction expiry. The base fee update is not treated as a separate event, it is performed as part of every block production event.

2.1 Transaction arrival

Transaction arrivals follow a Poisson process with rate λ. At the moment of arrival of a new transaction, three quantities are generated: the volume of consumed gas $g_n \sim LogNormal(\bar{g}, \sigma_g)$, maximum price for unit gas $f_n^{\max} \sim LogNormal(\bar{f}, \sigma_f)$, tip size $\pi_n \sim Exp(\bar{\pi})$.

A lognormal random variable with parameters $(\bar{x}, \sigma_x)$ satisfies

$$\ln X \sim \mathcal{N}(\bar{x}, \sigma_x^2).$$

2.2 Block production

The blocks appear with exponentially distributed intervals, the average is 12 seconds. Block processing consists of transaction selection and base fee adjustment.

Selection procedure is conducted only among eligible transactions, i.e. $f_n^{\max} \geq b_t$. They are sorted in descending order of tip and start to be included in the block. If the best candidate exceeds the remaining gas budget, we check for the next one until there are no eligible transactions left that haven't been considered.

2.3 Base Fee Update

The key feature of the EIP-1559 upgrade is the base fee update, which is realised through this formula:

$$b_{t+1} = b_t \left(1 + \frac{1}{8} \cdot \frac{G_{\text{used}} - G^*}{G^*}\right), \quad G^* = 15 \times 10^6$$

It is clear by inspecting the formula that the base fee increases for oversaturated blocks, and decreases for unfilled blocks proportionally to the relative deviation of block utilisation from the target value. We use a lower bound for stability $b_{t+1} = \max(b_{t+1}, 1)$

2.4 Transaction Expiry

After arrival each transaction is assigned an expiry after $T_{exp} = 300$ seconds. If the transaction is still in mempool by this time, it is marked as expired and gets removed. In case it was accepted, the expiry event is removed at the moment of confirmation.

3 Flowchart

4 Steady-state analysis

The metrics of interest are long-run averages. Their estimation requires handling two issues: the bias from the artificial start state (empty mempool, $b_0 = 10$ Gwei), and the correlation between successive observations within a single run.

4.1 Warm-up period

The warm-up length D is found through a Welch-type heuristic. We run N_W independent replications of length K_W and average the k-th confirmation time across replications:

$$\overline{W}_{\bullet k} = \frac{1}{N_W} \sum_{n=1}^{N_W} W_{nk}.$$

D is the smallest index ($D \geq 100$, $2D \leq K_W$) for which

$$\left| \frac{\frac{1}{2D} \sum_{k=1}^{2D} \overline{W}_{\bullet k}}{\frac{1}{D} \sum_{k=1}^{D} \overline{W}_{\bullet k}} - 1 \right| \leq 0.05.$$

This means that the average over the first $2D$ observations differs from the average over the first D by at most 5%. The first D confirmations are discarded, and time-weighted statistics are reset at this moment with the current mempool size as the starting value, so that the integral continues from the correct point.

4.2 Batch means

After warm-up, the sequence is split into r consecutive batches of b observations each, with batch averages

$$\overline{D}_\ell = \frac{1}{b} \sum_{k=(\ell-1)b+1}^{\ell b} D_k.$$

For large enough b, the batch means are approximately independent. We use $r = 30$ and $b = 4D$ according to the standard rule of thumb. The point estimate and 95% confidence interval are

$$\overline{\overline{D}}(r) = \frac{1}{r} \sum_{\ell=1}^{r} \overline{D}_\ell, \qquad \overline{\overline{D}}(r) \pm t_{r-1,\,1-\alpha/2} \, \frac{\sqrt{S^2(r)}}{\sqrt{r}},$$

with $S^2(r) = \frac{1}{r-1} \sum_\ell (\overline{D}_\ell - \overline{\overline{D}}(r))^2$.

4.3 Regenerative method

A regeneration point is a moment at which the future evolution of the system becomes probabilistically independent of the past. We use the production of a block that leaves the mempool empty as such a moment. At this point no transactions remain in the mempool, all expiry events have either fired or been cancelled, and the only pending events are the next arrival and the next block, both governed by memoryless exponential clocks. The base fee itself is not reset at this moment, but in steady state it concentrates around a stable value, so this approximation is acceptable.

For each cycle k, let W_k^{total} be the sum of confirmation times and M_k the number of confirmations in the cycle. The estimator is the ratio of means

$$\widehat{W} = \frac{\frac{1}{N} \sum_k W_k^{total}}{\frac{1}{N} \sum_k M_k}.$$

This is a ratio of two random variables, so the standard CLT does not apply directly. We use the transformation $V_k = W_k^{total} - \widehat{W} M_k$, which has zero mean in steady state. The 95% CI is then

$$\widehat{W} \pm t_{N-1,\,1-\alpha/2} \, \frac{\sqrt{\widehat{\mathrm{Var}}(V)/N}}{\overline{M}}.$$

Under heavy load, the mempool may rarely empty within a tractable run length. In that case too few cycles are observed and batch means is the only viable estimator.

4.4 Stopping rule

The run length is not fixed in advance. The simulation is extended until each reported metric satisfies the relative precision target $\delta = 0.10$: the half-width of its 95% CI must be at most 10% of the point estimate. If any metric fails, the number of observations is increased and the run is repeated.

4.5 Comparison Experiment

For the comparison experiment, we use the lognormal parameters specified in the assignment:

$$\sigma_g = 0.5, \qquad \sigma_f = 0.3,$$

$$\bar{g} = \ln(50000) - \frac{\sigma_g^2}{2} \approx 10.69, \qquad \bar{f} = \ln(100) - \frac{\sigma_f^2}{2} \approx 4.56.$$

The mean tip is set to

$$\bar{\pi} = 2.$$

We compare two transaction arrival rates:

$$\lambda = 8tx/s$$

and

$$\lambda = 12tx/s.$$

The first case represents normal load, while the second case represents a morecongested load. For both cases, the same EIP-1559 base-fee mechanism, gas-packing rule, and transaction expiry rule are used.

4.6 Results

Table 4.6 shows preliminary simulation results for the two load scenarios.

Metric	$\lambda = 8$ tx/s	$\lambda = 12$ tx/s
Simulation time	255.07 s	176.47 s
Total submitted	2054	2104
Total confirmed	2054	2104
Total expired	0	0
Average confirmation time	12.43 s	27.36 s
Average mempool size	100.06	326.24
Average ineligible fraction	0.0000	0.0000
Average block gas utilisation	0.1786	0.6903
Expiry rate	0.0000	0.0000
Final base fee	2.00 Gwei	12.44 Gwei

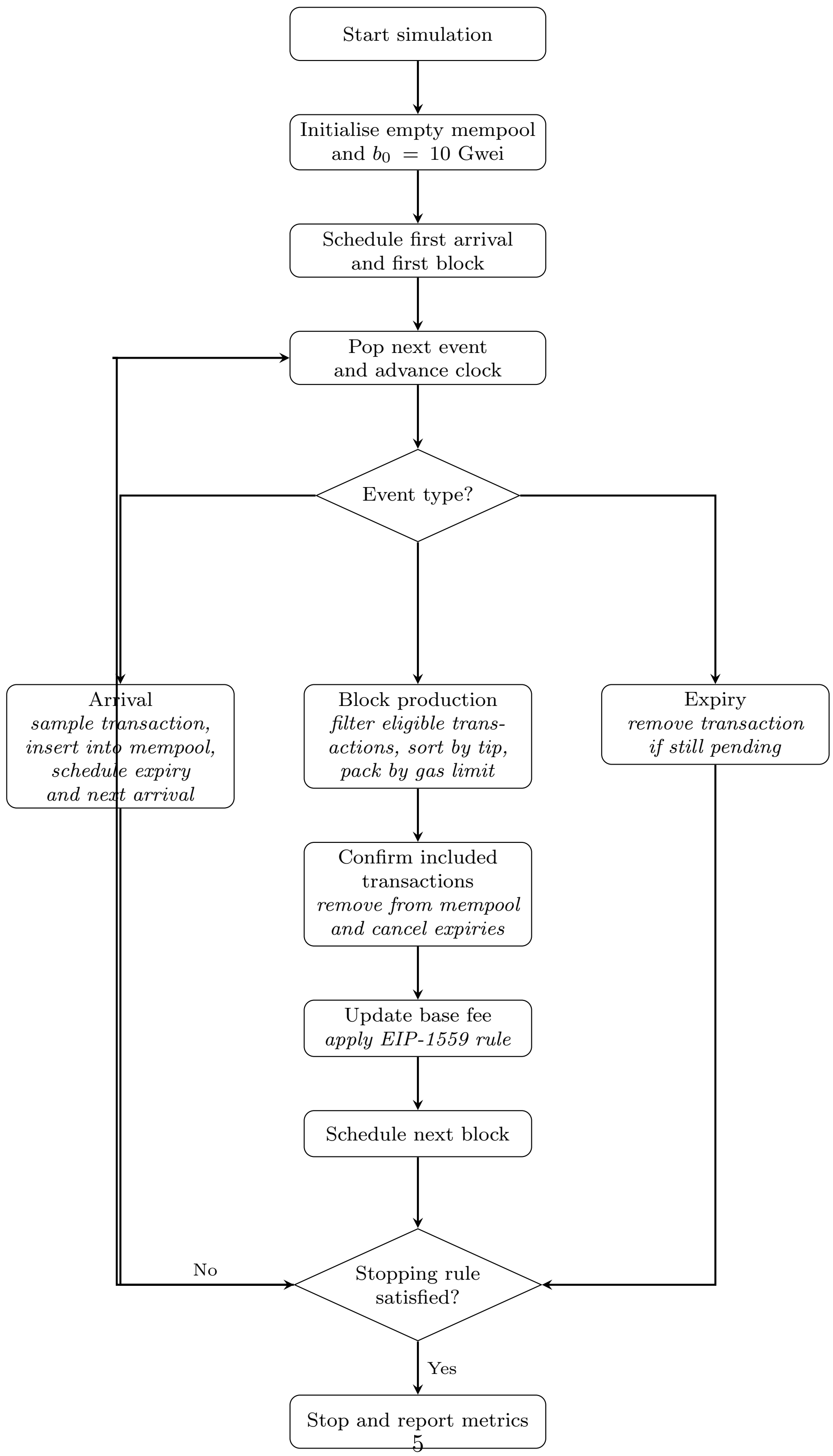

Figure 1: Flowchart of the EIP-1559 Ethereum mempool simulation. The simulation has three event types: transaction arrival, block production, and transaction expiry. The base fee update is performed inside the block production event.